10 **(a)** State what is meant by coherent light waves.

.....

..... [1]

(b) Coherent light of wavelength of 633 nm is incident normally on a pair of narrow parallel slits separated by a distance of 0.250 mm as shown in Figure 10.1.

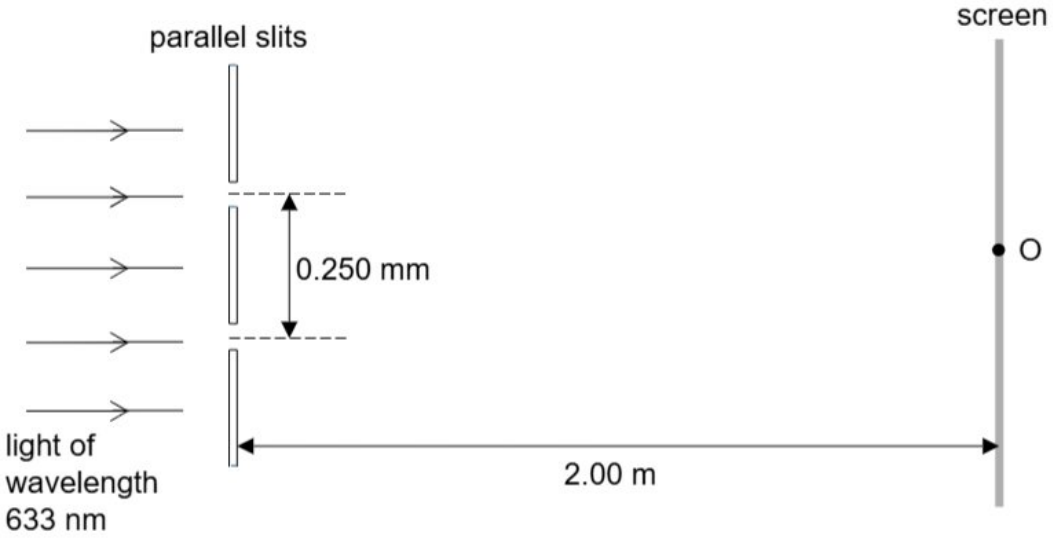


Figure 10.1 (not to scale)

An interference pattern of bright and dark fringes is seen on a screen placed parallel to the plane of the slits. The screen is 2.00 m away from the slits. Point O is equidistant from both slits.

(i) Explain how the bright and dark fringes are formed on the screen.

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..... [2]

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(ii) Calculate the distance between the central bright fringe and the second-order dark fringe on the screen.

distance = ... mm [2]

(iii) Explain why the maximum observed at O is of a greater intensity than any other observable maximum.

.....

.....

.....

.....

..... [2]

(c) The setup is modified by blocking one of the slits, leaving a single slit of width 0.080 mm exposed to the original light source of single wavelength of 633 nm as shown in Figure 10.2.

A diffraction pattern of the single slit is formed on the screen centred at point P.

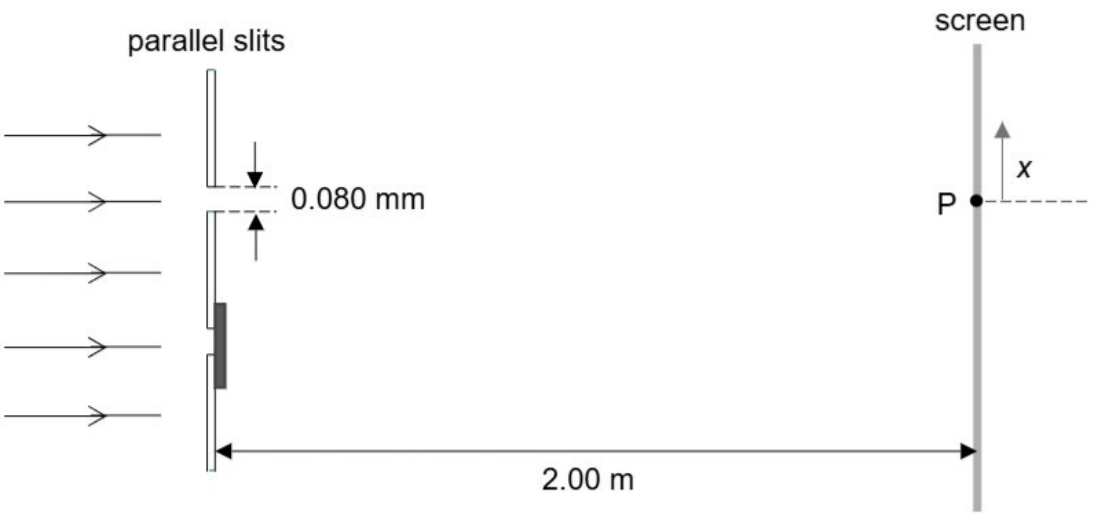


Figure 10.2 (not to scale)

(i) Show that the width of the central fringe on the screen is approximately 31.7 mm.

[2]

(ii) Part of the diffraction pattern showing the variation with displacement x from P of the intensity of light is shown in Figure 10.3.

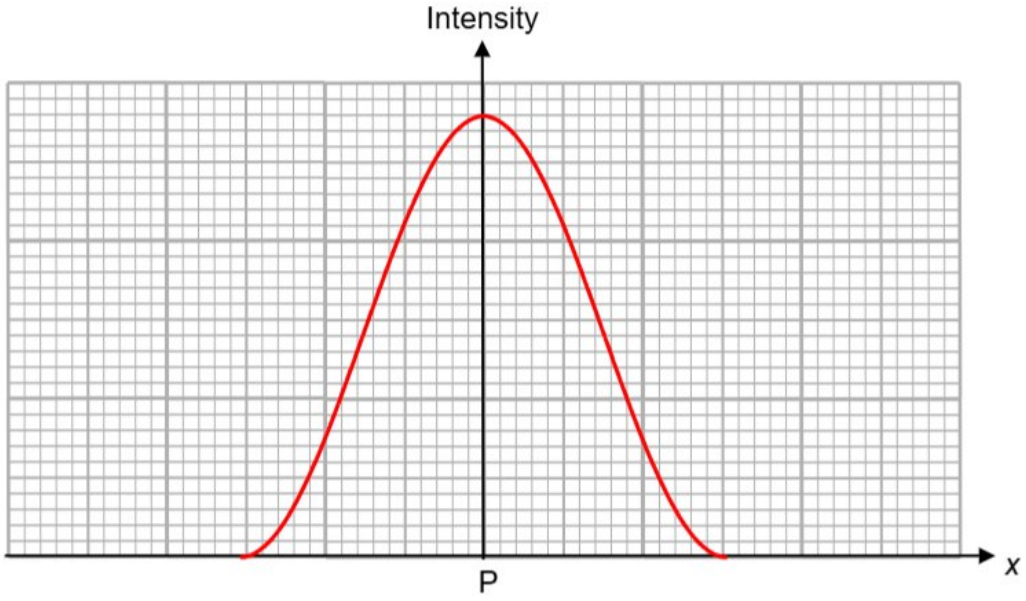


Figure 10.3

The slit width is reduced by half. Sketch on Figure 10.3 the new diffraction pattern. [2]

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(iii) The single slit is now used as an aperture for a simple optical imaging system.

Two small LED indicators are spaced 5.0 mm apart as shown in Figure 10.4. The LED indicators emit light of wavelength 633 nm.

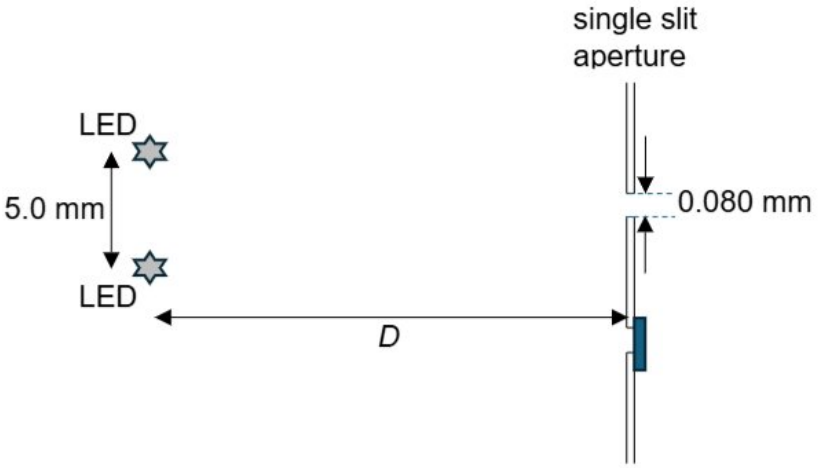


Figure 10.4 (not to scale)

Calculate the maximum distance *D* such that the images from the LED indicators can be just resolved.

D = ... m [3]

(d) A parallel beam of light of wavelength 633 nm is incident normally onto a 6.00 m screen. A bright spot is observed at the centre of the screen.

A diffraction grating is inserted normally into the path of the light as shown in Figure 10.5. The grating has 500 lines per millimetre.

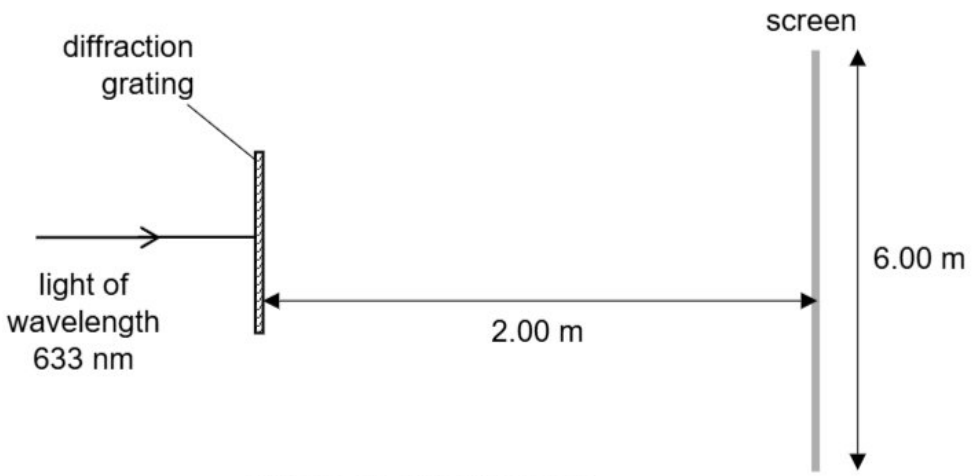


Figure 10.5 (not to scale)

(i) Determine the maximum number of bright lines that can be observed on the screen.

maximum number = ... [4]

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(ii) The incident beam is now changed to a source of white light.

State and explain how the appearance of the first-order spectrum differs from the central maximum.

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..... [2]

[Total: 20]

~ End of Paper ~